Mathematics for Business and Economics I

Prof. Vincent Ginis, Melika Mobini & Vincent Holst

Bonus Test

Student ID

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Version

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Disclaimer: Only answers in the black boxes will be graded. Write down intermediate steps for full points. If your student ID ends with an even number, make version A. Otherwise, make version B. Indicate your group and version left.

Question 1

$$\int \frac{2}{\sqrt{x}} - \frac{1}{x} + 3\,dx = 4\sqrt{x} - \ln(|x|) + 3x + C$$

for $C \in \mathbb{C}$

Question 2

$$\int \cos(3x)\,(\sin(3x))^2\,dx = \frac{1}{3}\int 3\cos(3x)\,(\sin(3x))^2\,dx$$

$$= \frac{1}{9}(\sin(3x))^3 + C$$

for $C \in \mathbb{C}$

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Question 1

$$= \frac{2}{7}x^{\frac{7}{2}} + \frac{x^2}{2} + x$$

Question 2

$$= x\ln(x) + \int dx$$

$$= x\ln(x) + x$$

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Question 1

$$\int \sin(x)^2 + \int \frac{1}{x^2} + 2x$$

$$2\int \sin(x) + \ln x^2 + 2x$$

$$2\cos x + \ln x^2 + 2x.$$

Question 2

$$u = \sqrt{x^2+4}$$

$$du = 2x\,dx$$

$$\int \frac{x}{u} \cdot \frac{du}{2x}$$

$$\int \frac{du}{u2} = \ln \frac{u}{2} = \frac{\ln\sqrt{x^2+4}}{2}$$

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Question 1

$$\int\left(\frac{5}{x^2}+\frac{1}{x}+1\right)dx = \int \frac{5}{x^2}dx + \int \frac{1}{x}dx + \int 1 dx$$

$$= 5\int x^{-2}dx + \int x^{-1}dx + x$$

$$= 5\,\frac{x^{-1}}{-1} + \ln|x| + x$$

$$= -\frac{5}{x} + \ln|x| + x$$

Question 2

$$\int x e^x dx$$

partial integration :

$$u = x \rightarrow du = dx$$

$$dv = e^x dx \rightarrow v = e^x$$

$$= xe^x - \int e^x dx$$

$$= xe^x - e^x$$

$$= (x-1)e^x$$

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Question 1

$\int -\cos(x) + \frac{1}{x^2} + 2 \, dx = -\int \cos(x)\,dx + \int \frac{dx}{x^2} + 2\int dx$

$= -\sin(x) + C_1 + \left(-\frac{1}{x}\right) + C_2 + 2x + C_3$

$= -\sin(x) - \frac{1}{x} + 2x + C$ as C_1, C_2, C_3 are arbitrary constants.

Question 2

$\int x\cos(x)\,dx = x\sin(x) - \int 1\cdot\sin(x)\,dx$

$= x\sin(x) - (-\cos(x)) + C$

$= x\sin(x) + \cos(x) + C$.

Partiële integratie:

Stel: $u = x$

$\Rightarrow du = 1\,dx$

$dv = \cos(x)\,dx$

$\Rightarrow v = \sin(x)$

$\int u\,dv = u\cdot v - \int v\,du$

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Question 1

$$\int \frac{5}{x^2} + \frac{1}{x} + 1 \, dx$$

$$\int \frac{1}{x} + \frac{5}{x^2} + 1 \, dx$$

$$\int 1 \, dx + \frac{5}{x^2} + \frac{1}{x}$$

Question 2

A) $$\int (2x+3) e^{x^2+3x} \, dx$$

$$\int (3x+4) e^{x^3} + 4x \, dx$$

$$\int (1+2x)^{ex^4} + 5x \, dx$$

B) $$\int x e^{x} \, dx$$

$$\int x e^{x^2} \, dx$$

$$\int 2x \, dx$$

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Question 1

$$\int \frac{5}{x^2} + \frac{1}{x} + 1 \, dx$$

$$= \int 5x^{-2} + x^{-1} + 1 \, dx$$

$$= -5x^{-1} + \ln(|x|) + C$$

Question 2

$$\int (2x+3)e^{x^2+3x} \, dx = \int u v'$$

$u = 2x+3$, $u' = 2$

$\frac{g(x)}{dx} = v' = e^{x^2+3x}$

$v = e^{x^2+3x}$

$$= (2x+3)(e^{x^2+3x}) - \int (e^{x^2+3x})2 + C$$

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Question 1

$$\int x^2\sqrt{x}\,dx + \int x\,dx + \int 1\,dx$$

$$= \int x^{\frac{5}{2}}\,dx + \frac{x^2}{2} + x + C'$$

$$\cancel{= x^2 + x + C}$$

$$\cancel{= x(x+1) + C}$$

$$= \frac{2x^{\frac{5}{2}}}{5} + \frac{x^2}{2} + x + C'$$

Question 2

$$\int \frac{3x}{(2x^2+1)^2}\,dx = \int \frac{3\,dx^2}{2(2x^2+1)^2} = \int \frac{3}{4(2x^2+1)^2}\,d(2x^2+1)$$

$$= \frac{3}{4}\int (2x^2+1)^{-2}\,d(2x^2+1)$$

$$= \frac{3}{4}\cdot -1\,(2x^2+1)^{-1} + C'$$

$$= \frac{-3}{4}\,\frac{1}{(2x^2+1)} + C'$$

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Question 1

$$\int \frac{x^3}{2} + \sin(x) + 1 \, dx$$

$$= \frac{1}{4}x^4 + \cos(x) + x + C$$

Question 2

$$\int \frac{4x+6}{2x^2+6x+5} \, dx$$

$$u = 2x^2+6x+5$$

$$\frac{du}{dx} = 4x+6$$

$$= \int \frac{\cancel{4x+6}}{u} \frac{du}{\cancel{4x+6}}$$

$$= \int \frac{1}{u} \, du = \ln(u) + C$$

$$= \ln(2x^2+6x+5) + C$$

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4	○	○	○	○	○	○	○
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Question 1

$$\int \frac{x^3}{2} + \sin(x) + 1 \, dx$$

$$= \frac{3}{2}x^2 + \cos(x) + C$$

Question 2

$$I = \int (2+5x)\, e^{\frac{x}{3}} \, dx$$

$u = e^{\frac{x}{3}}$ $\quad v = 5$

$u' = \frac{1}{3} e^{\frac{x}{3}}$ $\quad v' = 2+5x$

$$I = 5e^{\frac{x}{3}} - \int \frac{5}{3} e^{\frac{x}{3}}$$

$$= 5e^{\frac{x}{3}} - \frac{5}{3} e^{\frac{x}{3}} = 0 + C$$